The leximin core selection on exact games

Draft fragment: singleton theorem, exact bump paths, and the open crux

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July 2026 — working fragment

Status: mathematical core of a paper in progress. Singleton bumps are proved monotone even on balanced games; grand-coalition bumps are proved to remain on an exact path, with labeled resource monotonicity still open. Proper nonsingleton bumps are reduced to an exact capacity dichotomy and one open path lemma. Balanced nonexact counterexamples establish that exactness is necessary.

1 The leximin-above-floors core selection

Let (N, v) be a balanced TU game, $M = v(N)$, and floors $m_i := v(\{i\})$. For $x \in \mathbb{R}^N$ write the *gains* $g_i(x) = x_i - m_i$. The *leximin-above-floors* selection is

$$\phi(v) := \arg \operatorname*{leximin}_{x \in \mathrm{Core}(v)} \big(g_1(x), \ldots, g_n(x)\big),$$

the unique core allocation whose ascending-sorted gain vector is lexicographically largest. It is single-valued (leximin over a polytope) and is computed by the Maschler-type scheme over player coordinates: repeatedly raise a common lower level s of the free players' gains as far as the core allows, fix ("pin") every free player whose gain cannot exceed $m_i + s$ on the level-optimal face, and recurse on the rest. On *exact* games $m_i = \min_{\mathrm{Core}} x_i$, so $g_i \geq 0$ and ϕ egalitarianly distributes the surplus $x_i - v(\{i\})$ inside the core; on convex games it is the constrained egalitarian solution of Dutta and Ray (1989). Implementation (exact rational + float) in `src/leximin.py`, validated against those two coincidences.

We ask whether ϕ is *coalitionally monotonic*: if $v' = v + d\,\mathbf{1}_S$ for a single coalition S and $d > 0$, is $\phi_i(v') \geq \phi_i(v)$ for every $i \in S$? A positive answer on exact games would exhibit an explicit coalitionally monotonic core selection on that class. The reduction in Section 5 addresses only proper nonsingleton coalitions. Singleton bumps change the floors defining ϕ, and a grand-coalition bump changes the efficiency hyperplane. The singleton case is resolved next; the grand-coalition case requires a separate resource-monotonicity argument and remains open in this fragment.

2 Singleton bumps are monotone

We first isolate an elementary property of the leximin order. Write $a \succ_{\mathrm{lx}} b$ when the ascending rearrangement of a is lexicographically larger than that of b.

Lemma 2.1 (one-coordinate translation)**.** *If $a \succ_{\mathrm{lx}} b$, $a_i < b_i$, and $d > 0$, then $a + de_i \succ_{\mathrm{lx}} b + de_i$.*

Proof. For a finite vector z, put $L_\lambda(z) = \sum_j e^{-\lambda z_j}$. The first unequal coordinate of the ascending rearrangements dominates this sum as $\lambda \to \infty$; hence $a \succ_{\mathrm{lx}} b$ if and only if $L_\lambda(a) < L_\lambda(b)$ for all sufficiently large λ. Moreover,

$$L_\lambda(b + de_i) - L_\lambda(a + de_i) = L_\lambda(b) - L_\lambda(a) + (1 - e^{-\lambda d})(e^{-\lambda a_i} - e^{-\lambda b_i}).$$

Both terms on the right are positive for all sufficiently large λ, because $a_i < b_i$. The same characterization gives the claim. □

Theorem 2.2 (singleton coalitional monotonicity)**.** *Let v and $v' = v + d\,\mathbf{1}_{\{i\}}$ be balanced games, with $d > 0$. Then*

$$\phi_i(v') \geq \phi_i(v).$$

Thus singleton bumps are monotone even without exactness.

Proof. Let $m_j = v(\{j\})$ and express the old core in gain coordinates

$$G = \{(x_j - m_j)_{j \in N} : x \in \mathrm{Core}(v)\}.$$

If the singleton floor of player i is raised by d, the new feasible gain set is

$$H = \{g - de_i : g \in G,\ g_i \geq d\}.$$

Let z be the leximin point of G and h the leximin point of H. The desired inequality is $h_i + d \geq z_i$. If $z_i < d$, this is immediate from $h_i \geq 0$. Suppose $z_i \geq d$. Then $b = z - de_i$ belongs to H, so h is either b or $h \succ_{\mathrm{lx}} b$. If, contrary to the claim, $h_i + d < z_i$, then $h_i < b_i$ and the equality case is impossible. Lemma 2.1 gives

$$h + de_i \succ_{\mathrm{lx}} b + de_i = z.$$

But $h + de_i \in G$, contradicting the definition of z as the leximin point of G. □

3 Grand-coalition bumps stay exact

The grand-coalition case also has an exact path, but its geometry differs from the nested-core path of Section 5.

Lemma 3.1 (grand-bump exact path)**.** *Let v be exact and, for $r \geq 0$, let v^r agree with v on every proper coalition and satisfy $v^r(N) = v(N) + r$. Then v^r is exact for every $r \geq 0$. Moreover,*

$$\mathrm{Core}(v) + r\Delta_N \subseteq \mathrm{Core}(v^r), \qquad \Delta_N = \{a \in \mathbb{R}^N_+ : a(N) = 1\}.$$

Proof. If $x \in \mathrm{Core}(v)$ and $a \in r\Delta_N$, then $x + a$ has total $v(N) + r$ and weakly raises every coalition sum, proving the inclusion and balancedness. Fix a proper coalition T. Exactness of v supplies $x^T \in \mathrm{Core}(v)$ with $x^T(T) = v(T)$. Choose $j \notin T$. Then $x^T + re_j \in \mathrm{Core}(v^r)$ by the inclusion and remains tight at T. Thus

$$\min_{x \in \mathrm{Core}(v^r)} x(T) = v(T) = v^r(T).$$

The grand coalition is tight throughout its core by efficiency, so v^r is exact at every coalition. □

Lemma 3.1 removes exactness preservation from the grand case entirely: the remaining question is the resource-monotonicity statement

$$r \longmapsto \phi_i(v^r) \quad \text{is nondecreasing for every } i.$$

The inclusion in the lemma supplies the feasible monotone comparison point $\phi(v) + ra$ for every $a \in \Delta_N$, but leximin dominance of such points does not by itself imply coordinatewise dominance with player labels. Thus the unresolved content is a labeled resource-monotonicity theorem for these exact-core slices, not another bump-exactness lemma.

4 Exactness is necessary

Coalitional monotonicity of ϕ fails on balanced games in general—already at $n = 4$, and even at $n = 5$ where every floor is core-attained. So exactness cannot be weakened to balancedness.

Theorem 4.1. *There is a balanced (non-exact) five-player game v, a coalition S with $|S| = 3$, and $d > 0$ with $v + d\,\mathbf{1}_S$ balanced, such that some $i \in S$ has $\phi_i(v + d\,\mathbf{1}_S) < \phi_i(v)$. The same holds for a four-player game.*

Example 4.2 ($n = 5$; floors attained)**.** *Let $N = \{1, \dots, 5\}$ with $v(N) = 7$ and (all other coalitions as listed in the supplement) in particular*

$$v(\{1,2,3\}) = 3, \quad v(\{2,3\}) = 3, \quad v(\{1,3,5\}) = v(\{3,5\}) = 4, \quad v(\{1,5\}) = -1.$$

Floors are $m = (0,1,1,1,2)$, and $m_i = \min_{\mathrm{Core}} x_i$ for every *i (all floors attained); the game is balanced but not exact (e.g. $\min_{\mathrm{Core}} x(\{1,5\}) = 1 > -1$). Then*

$$\phi(v) = \left(\tfrac{1}{2}, \tfrac{3}{2}, \tfrac{3}{2}, 1, \tfrac{5}{2}\right), \qquad \phi(v + \mathbf{1}_{\{1,2,3\}}) = (1,1,2,1,2),$$

both in their respective cores, and player $2 \in \{1,2,3\}$ drops *from $\frac{3}{2}$ to 1 when $v(\{1,2,3\})$ is raised from 3 to 4. Thus exactness at the singletons (floor attainment) does not suffice: it is exactness at higher coalitions that the positive theorem needs.*

Example 4.3 ($n = 4$; floor unattained)**.** *Let $N = \{1,2,3,4\}$, $v(N) = 20$, with $v(\{1\}) = -3$, $v(\{1,2\}) = 11$, $v(\{1,3\}) = 7$, $v(\{2,3\}) = 4$, $v(\{1,2,3\}) = 13$, $v(\{2,4\}) = 10$, $v(\{1,3,4\}) = 9$, $v(\{2,3,4\}) = 13$, and the remaining values in the supplement. Here $v(\{1\}) = -3 < 1 = \min_{\mathrm{Core}} x_1$. Then $\phi(v) = (\frac{5}{2}, \frac{17}{2}, \frac{9}{2}, \frac{9}{2})$ and $\phi(v + 3\,\mathbf{1}_{\{1,2,3\}}) = (2,9,5,4)$: player $1 \in \{1,2,3\}$ drops from $\frac{5}{2}$ to 2.*

Both examples are certified in exact rational arithmetic (ϕ in the core on both sides, the drop exact); see the supplement and `src/hunt_leximin_nonexact.py`.

5 Reduction for proper nonsingleton bumps on exact games

Fix an exact v, a proper coalition S with $|S| \geq 2$, and $d > 0$ with $v' = v + d\,\mathbf{1}_S$ exact. Since $|S| \geq 2$ the floors are unchanged, and

$$\mathrm{Core}(v') = \mathrm{Core}(v) \cap \{x : x(S) \geq v(S) + d\} \subseteq \mathrm{Core}(v).$$

For $\tau \in [v(S), v(S)+d]$ let v_τ agree with v except $v_\tau(S) = \tau$, so $\mathrm{Core}(v_\tau) = \mathrm{Core}(v) \cap \{x(S) \geq \tau\}$ interpolates $\mathrm{Core}(v)$ and $\mathrm{Core}(v')$.

Lemma 5.1 (exact interval)**.** *Every v_τ, $\tau \in [v(S), v(S)+d]$, is exact.*

Proof. $\mathrm{Core}(v_\tau) \supseteq \mathrm{Core}(v') \neq \emptyset$, so v_τ is balanced. Exactness needs $\min_{\mathrm{Core}(v_\tau)} x(T) = v_\tau(T)$ for all T. For $T = S$, exactness of v supplies $x^0 \in \mathrm{Core}(v)$ with $x^0(S) = v(S)$, while any $x^1 \in \mathrm{Core}(v')$ satisfies $x^1(S) \geq v(S) + d$. The core of v is convex, so the segment $[x^0, x^1]$ contains a point $x^\tau \in \mathrm{Core}(v)$ with $x^\tau(S) = \tau$. This point belongs to $\mathrm{Core}(v_\tau)$, proving $\min_{\mathrm{Core}(v_\tau)} x(S) = \tau$. For $T \neq S$: put $f_T(\tau) = \min_{\mathrm{Core}(v_\tau)} x(T)$. As τ grows $\mathrm{Core}(v_\tau)$ shrinks, so f_T is nondecreasing; and $\mathrm{Core}(v_\tau) \subseteq \mathrm{Core}(v)$ gives $f_T(\tau) \geq \min_{\mathrm{Core}(v)} x(T) = v(T)$. At the endpoints $f_T = v(T)$ because v and v' are exact. A nondecreasing function equal to $v(T)$ at both endpoints and $\geq v(T)$ between is $\equiv v(T)$. Hence $f_T(\tau) = v(T) = v_\tau(T)$. □

The entire exact-bump range can be read from exposed core faces. For $T \subseteq N$, write $F_T = \{x \in \mathrm{Core}(v) : x(T) = v(T)\}$ (with $F_N = \mathrm{Core}(v)$), and define

$$d_v^*(S) = \min_{T \neq S} \left(\max_{x \in F_T} x(S) - v(S) \right).$$

Proposition 5.2 (exact bump capacity)**.** *For an exact game v and a proper coalition S, the bumped game $v + d\mathbf{1}_S$ is exact if and only if $0 \leq d \leq d_v^*(S)$. If $\sigma_S = \phi(v)(S) - v(S)$, then every exact bump at S is non-binding if and only if $d_v^*(S) \leq \sigma_S$; a binding exact bump exists if and only if $d_v^*(S) > \sigma_S$.*

Proof. The bumped core is $K_d = \mathrm{Core}(v) \cap \{x : x(S) \geq v(S) + d\}$. For $T \neq S$, exactness at T is equivalent to $F_T \cap K_d \neq \emptyset$, which is equivalent to

$$d \leq \max_{x \in F_T} x(S) - v(S).$$

Taking the minimum over $T \neq S$ gives all the higher-coalition exactness conditions; the term $T = N$ also gives nonemptiness of K_d. Exactness at S follows by interpolating between a point of F_S and a point of K_d, as in Lemma 5.1. This proves the first statement. The last two statements compare the exact capacity with the largest bump still satisfied by the old selection, namely σ_S. □

So the bump is realized by a continuous path of *exact* games in which only the lower bound τ on $x(S)$ rises. This isolates the role of exactness: if endpoint exactness is weakened to balancedness, the analogous interior path need not be exact, and that is precisely the room inhabited by the counterexamples of Section 4. For proper nonsingleton bumps on exact games, coalitional monotonicity is therefore equivalent to the one-parameter statement:

Lemma 5.3 (monotone path — OPEN)**.** *Along the exact path v_τ, the map $\tau \mapsto \phi_i(v_\tau)$ is nondecreasing for every $i \in S$ (and, empirically, nonincreasing for $i \notin S$).*

Two reductions cut the work down further. First, the common case is trivial:

Lemma 5.4 (non-binding)**.** *If $\phi(v)(S) \geq v(S) + d$ then $\phi(v') = \phi(v)$.*

Proof. $\phi(v)$ is the leximin point of $\mathrm{Core}(v) \supseteq \mathrm{Core}(v')$ and lies in $\mathrm{Core}(v')$, hence is the leximin point of the smaller set too; ϕ is single-valued. □

Binding ($\phi(v)(S) < v(S) + d$) requires $\phi(v)(S) < \max_{\mathrm{Core}} x(S)$ and is empirically rare. In every binding case observed (thousands of exact-preserving bumps at $n \leq 6$, zero violations) one finds $\phi(v')(S) = v(S) + d$ and the two-sided pattern $\phi_i(v') \geq \phi_i(v)$ for $i \in S$, $\leq$ for $i \notin S$.

Table 1: Computational evidence (float LP, exact-certifiable) for the reduction. $(\star_S)$ evaluated at ϕ of the base game; "binding" bumps only.

statement	sample	violations
ϕ monotone for sampled proper exact bumps (mixed sizes)	2000+ bumps, $n \le 6$	0
(D) exact $\Rightarrow (\star_S)$ at ϕ	39/39, 13/13	0
monotone $\Rightarrow (\star_S)$	all balanced	0
$(\star_S)$ along the path $\Rightarrow$ monotone	600	0
$(\star_S)$ at one endpoint $\Rightarrow$ monotone	1696 balanced	22 non-exact

A cone criterion. For $x = \phi(w)$ with tight set $\mathcal{T}(x) = \{T : x(T) = w(T)\}$, say $(\star_S)$ holds if there is $e \in \mathbb{R}^N$ with

$$e(N) = 0, \quad e(S) > 0, \quad e_i \ge 0 \ \ (i \in S), \quad e(T) \ge 0 \ \ (T \in \mathcal{T}(x)) :$$

a first-order direction raising $x(S)$ without lowering any member of S or breaking a tight core constraint. Empirically $(\star_S)$ is *necessary* for the bump to be monotone (it fails at $\phi(v)$ in Example 4.2, which is why player 2 must drop), and exactness supplies it. This suggests the skeleton

(A) Lemma 5.1: the bump path is exact;

(D) each exact v_τ satisfies $(\star_S)$ at $\phi(v_\tau)$;

(E′) $(\star_S)$ along the whole path implies Lemma 5.3,

whose composition is the theorem. The supporting evidence is summarized in Table 1; (D) and (E′) remain open, and $(\star_S)$ alone is *not* sufficient (the last row), so a proof of (E′) must use exactness along the whole path, not merely $(\star_S)$ at one endpoint.

The open problem, sharply. A Farkas dual turns (D) into a statement about a single maximal face of the core. For a core point y written as a mixture of core points, $y(T) \ge v(T) = \phi(T)$ at every tight T, so $e = y - \phi$ meets the tight-cone part of $(\star_S)$ automatically; hence

$$\begin{aligned}(\star_S) &\iff \exists\, y \in \mathrm{Core}(v) : \ y_i \ge \phi_i \ (i \in S) \text{ and } y(S) > \phi(S) \\ &\iff \max\{y(S) : y \in \mathrm{Core}(v),\, y_i \ge \phi_i \ (i \in S)\} > \phi(S).\end{aligned}$$

Dualizing (verified in 200/200 random balanced cases, exact-certifiable):

$(\star_S)$ *fails* at $\phi(v)$ if and only if $\phi(v)$ maximizes

$$\sum_{k \in S} r_k \, x_k$$

over $\mathrm{Core}(v)$ for some weights $r_k \ge 1$ on S and zero weights outside S.

Equivalently, the tight coalitions $\{T : \phi(T) = v(T)\}$ carry nonnegative weights and a $\beta \ge 1$ covering each $k \notin S$ exactly β and each $k \in S$ at most $\beta - 1$. Thus **Lemma 5.3's hypothesis (D) is exactly the assertion that an exact game's leximin point never lies on a**

positive-S-weighted maximal face of its core at a binding-bumpable S. Uniform weights $r_k \equiv c$ are already excluded, since maximizing $c\,x(S)$ would force $\phi(S) = \max_{\mathrm{Core}} x(S)$, contrary to binding-bumpability; *the entire open content is that exactness excludes the non-uniform such faces.* The non-exact counterexamples of Section 4 realize exactly such faces—this is why $(\star_S)$ fails there and a member of S drops. (A corner scan for a deficient S-member pinned at its core-maximum found no occurrence in any exact binding case, $0/16^{+}$.)

The natural certificate is provably wrong. Where $(\star_S)$ *does* fail, one would like to prove $d_v^*(S) = 0$ and thereby certify directly that S is not exactly bumpable. This target is false. From the failure certificate, take the cover $\{\lambda_U\}$ dual to the failure, $W = \bigcup\{U : \lambda_U > 0\}$, and propose $C = W \cap S$. This route is closed by a six-player exact game (supplement, `WITNESS_KEY_FALSE_n6`): $(\star_S)$ fails at $S = \{1,3,4,5\}$ with cover $\{2,3\},\{1,2,6\},\{3,4,6\}$, yet a core point x with $x(W \cap S) = v(W \cap S)$ attains $x(S) = 24 > \phi(S) = 71/3$, so $W \cap S$ certifies nothing; the associated slack inequality $\phi(W) - v(W) \le \phi(W \cap S) - v(W \cap S)$ fails by $1/3$. Such failures are rare ($\approx 0.1\%$ of $(\star_S)$-failures at $n \in \{6,7\}$, found only under generators biased toward thin cores with nested tight chains), which is why sampling first suggested the cover certificate was universal. In every observed failure, however, the operative bound survives: there is a coalition C with $M^{(C)} := \max\{x(S) : x \in \mathrm{Core}(v),\, x(C) = v(C)\} \le \phi(S)$, so that S cannot be bumped past $\phi(S)$ while preserving exactness (in a $\approx 0.2\%$ minority $d_v^*(S) > 0$ — exact six-player witnesses with $d^* = 1$ and $d^* = 2$ are included in the supplement — but always $d_v^*(S) \le \phi(S) - v(S)$). Empirically the certificate has sharp structure: C can always be taken as a subset of S contained in a *single stratum* of the leximin computation — almost always a singleton — and it is provably never ϕ-tight (a ϕ-tight certificate would force $\phi(S) = v(S)$, contradicting positive slack). The open problem is thus sharpened: produce, from the structure of a $(\star_S)$-failure at the leximin point of an exact game, a single-stratum $C \subseteq S$ with $M^{(C)} \le \phi(S)$. Empirically the certificate is pinned further: writing r for the residuals of the failure's dual cover ($r_i \ge 1$ on S), a *minimum-residual* member $k \in S$ (one with $r_k = 1$) certifies in 99.6% ($n = 6$) to 100% ($n = 7$) of observed failures, the residue requiring a same-stratum pair. By strong duality the singleton form is equivalent to exhibiting nonnegative weights on ϕ-tight coalitions with coverage β on $N \setminus S$, $\beta - 1$ on $S \setminus \{k\}$, and at least $\beta - 1$ on k; weights supported inside S suffice in only about two thirds of cases, so a proof must construct this dual cover from ϕ-tight coalitions *crossing* S. Equivalently, for a minimum-residual k the open claim reduces to the single inequality $\sum_{i \in S \setminus \{k\}} (r_i - 1)\,(x_i - \phi_i) \le 0$ for every core x with $x_k = v(\{k\})$: flooring the most-covered member must leave the weighted displacement of the less-covered members of S at or below zero relative to their leximin values. This inequality — which the residual weights alone do not force — is the precise frontier at which we leave the problem.

Proposition 5.2 removes the frequency language from the logical frontier. Every pair (v, S) falls into the exact dichotomy

$$d_v^*(S) \le \sigma_S \quad \text{or} \quad d_v^*(S) > \sigma_S.$$

The first side admits no binding exact bump and is finished by Lemma 5.4. On the second side a binding exact bump exists, and the missing structural statement is precisely

$$d_v^*(S) > \sigma_S \quad \Longrightarrow \quad (\star_S).$$

Thus the singleton/pair certificate statistics motivate how to prove this implication, but neither their observed percentages nor sampling rarity enters the conditional theorem.

Remark 5.5. *Lemma 5.3 is a leximin parametric-monotonicity statement; its cousin the nucleolus is* not *coalitionally monotone, so exactness (Lemma 5.1) is doing real work and the "self-defeating within-S transfer" heuristic is not by itself a proof. We leave (D), (E′) open and record the conditional theorem:* on exact games, ϕ is monotone under proper nonsingleton bumps provided Lemma 5.3 holds*; singleton and grand-coalition bumps remain outside this reduction, but the singleton case is settled unconditionally by Theorem 2.2; the necessity direction (Theorem 4.1) is unconditional.*

References

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